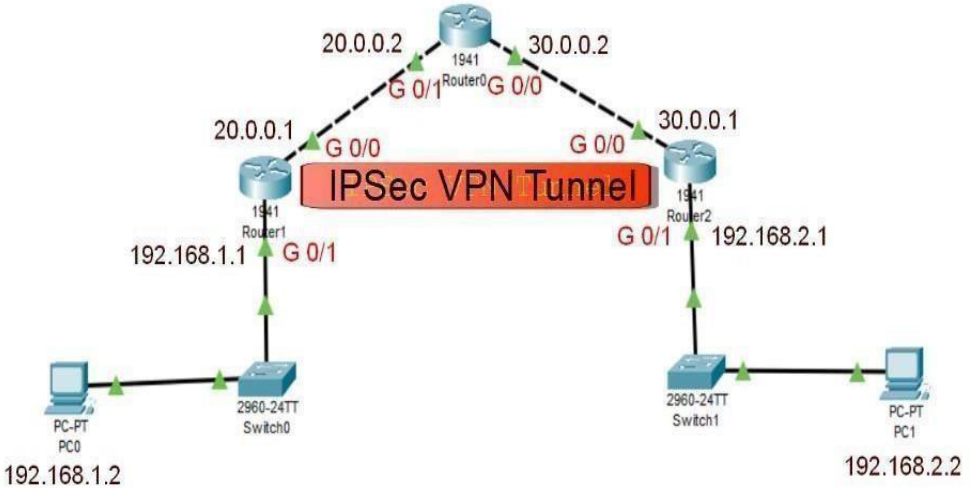
Practical No. 8

Aim: IP Security (IPsec)

Configuration:

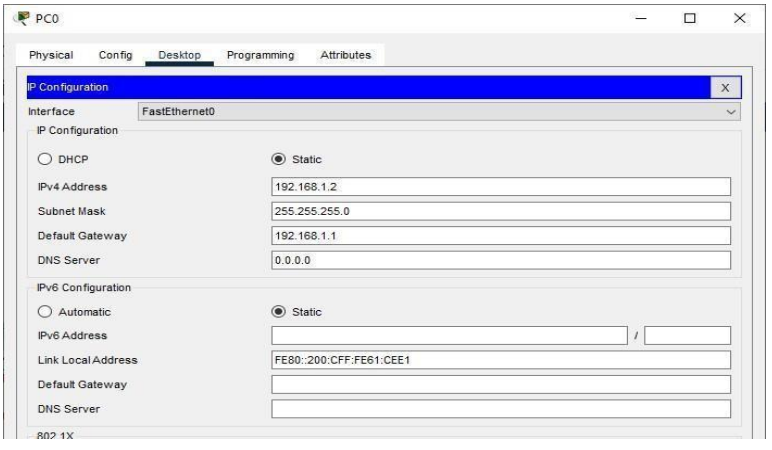
Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.



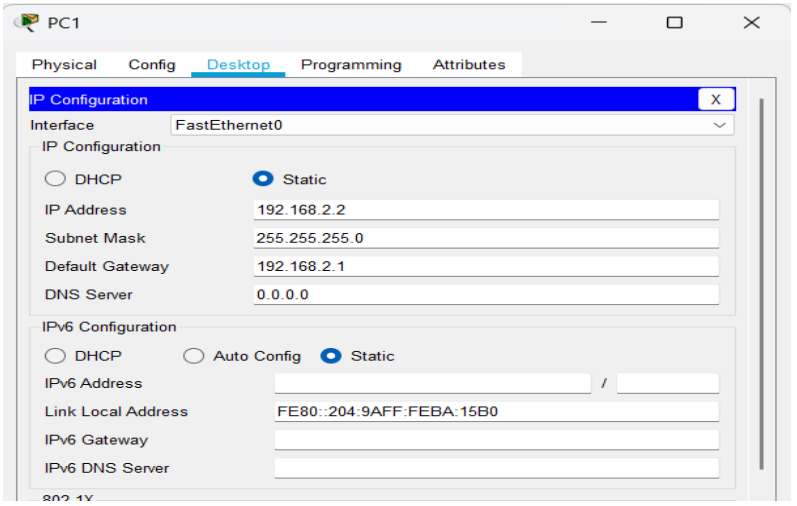
IPSec Network Addressing Table

Device	Interface	IP Address	Subnet Mask	Remarks
R0	g0/0	30.0.0.2	255.255.255.252	Link to R2
R0	g0/1	20.0.0.2	255.255.255.252	Link to R1
R1	g0/0	20.0.0.1	255.255.255.252	Link to R0
R1	g0/1	192.168.1.1	255.255.255.0	LAN (PC1)
R2	g0/0	30.0.0.1	255.255.255.252	Link to R0
R2	g0/1	192.168.2.1	255.255.255.0	LAN (PC2)
PC1	NIC	192.168.1.2	255.255.255.0	Connected to R1
PC2	NIC	192.168.2.2	255.255.255.0	Connected to R2

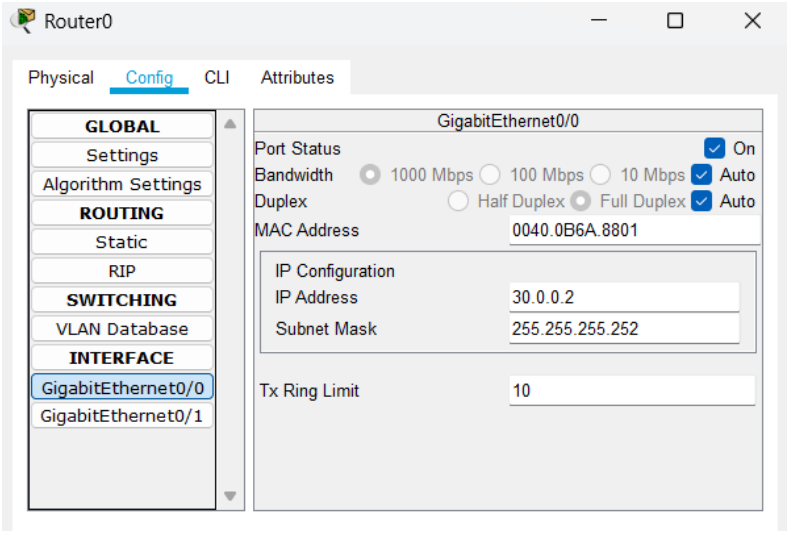
1. Configuring PC0:



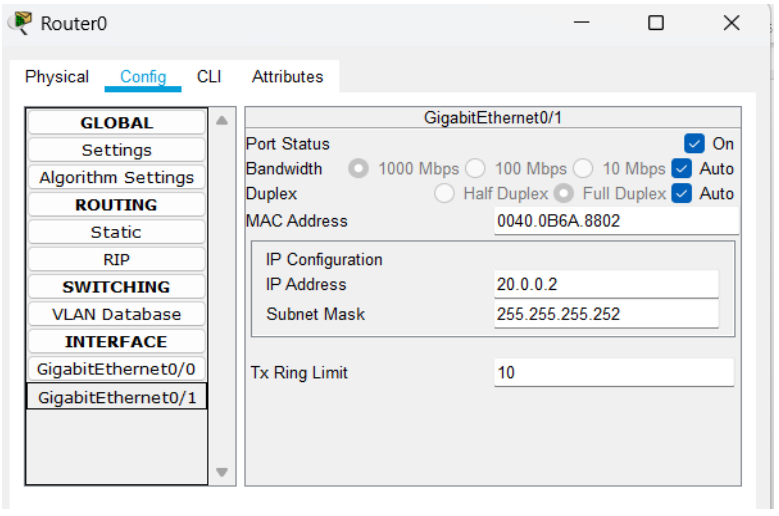
2. Configuring PC1:



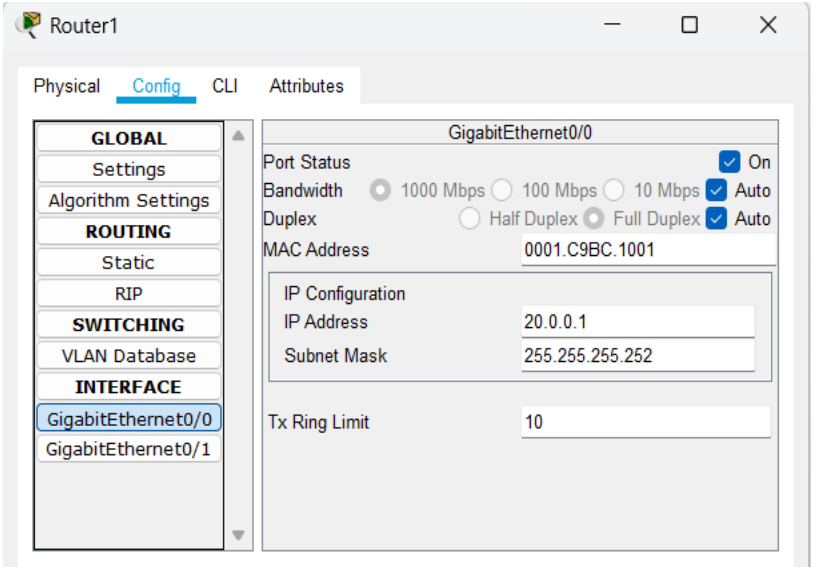
- **Configuring Router0:**
    1. **Interface GigabitEthernet0/1:**



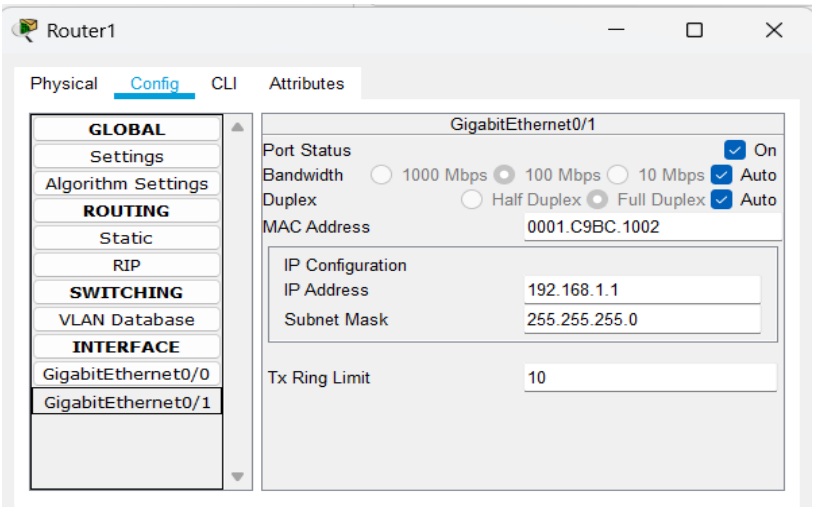
    2. **Interface GigabitEthernet0/0:**



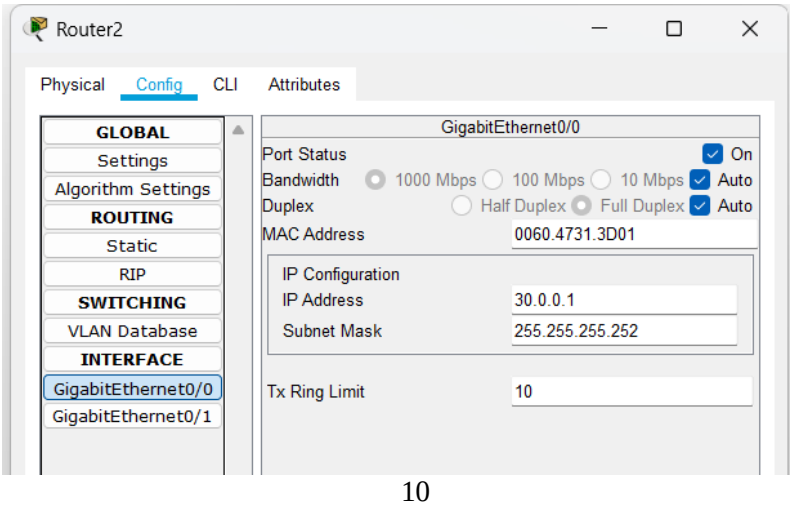
- **Configuring Router1:**
    1. **Interface GigabitEthernet0/0:**



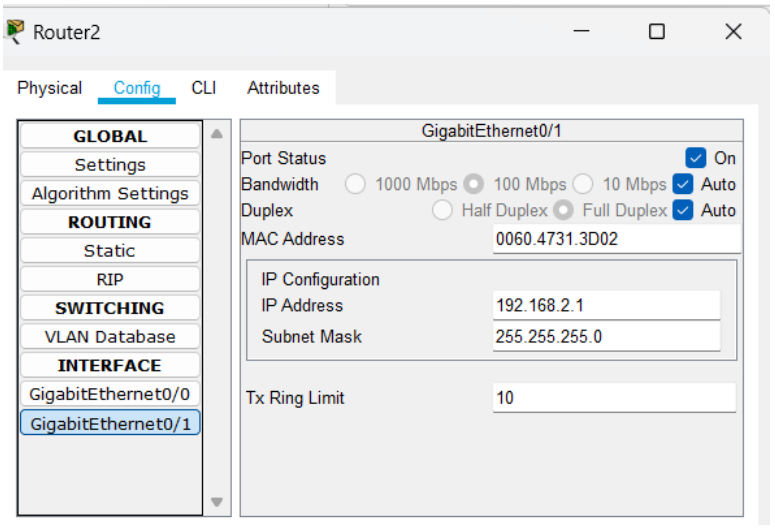
    2. **Interface GigabitEthernet0/1:**



- **Configuring Router2:**
    1. **Interface GigabitEthernet0/0:**



2. **Interface GigabitEthernet0/1:**



Checking and Enabling the Security features in Router R1 and R2:

1. **Enter the following command in the CLI mode of Router1**

Router(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
Router(config)#hostname R1
R1(config)#exit
R1#show version

```
Device#   PID                     SN
------------------------------------------------
*0        CISCO1941/K9            FTX1524N826-


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      None          None          None
data          None          None          None

Configuration register is 0x2102
```

(We see that the security feature is not enabled, hence we need to enable the security packageR1#)

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#license boot module c1900 technology-package securityk9
R1(config)#exit
R1#
R1#copy run startup-config
R1#reload
R1>enable

R1#show version

(The security package is enabled)

```
Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology     Technology-package          Technology-package
               Current        Type         Next reboot
-----------------------------------------------------------------
ipbase         ipbasek9       Permanent    ipbasek9
security       securityk9     Evaluation   securityk9
data           disable        None         None

Configuration register is 0x2102
```

2. Enter the following command in the CLI mode of Router2

Router(config)#ip route 0.0.0.0 0.0.0.0 30.0.0.2
Router(config)#hostname R2
R2(config)#exit
R2#show version

```
Device#   PID                   SN
------------------------------------------
*0        CISCO1941/K9          FTX1524N826-

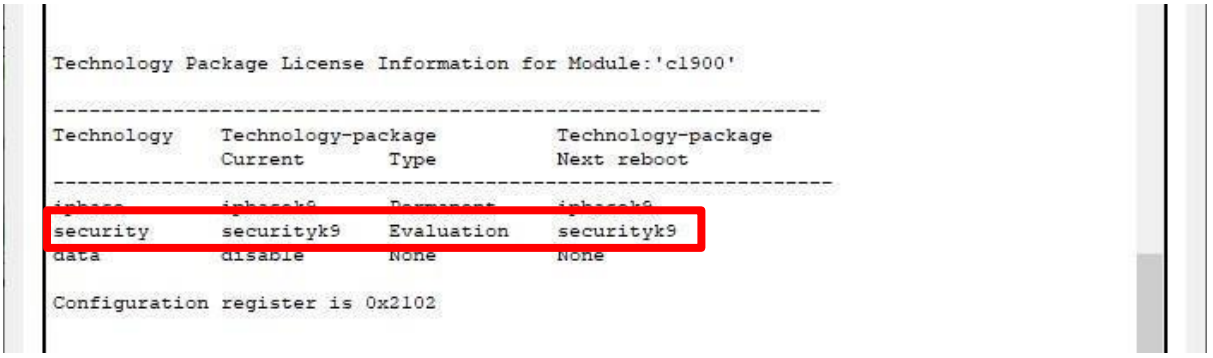

Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology     Technology-package          Technology-package
               Current        Type         Next reboot
-----------------------------------------------------------------
ipbase         ipbasek9       Permanent    ipbasek9
security       None           None         None
data           None           None         None

Configuration register is 0x2102
```

(We see that the security feature is not enabled, hence we need to enable the security packageR2#)

R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)# license boot module c1900 technology-package securityk9
R2(config)#exit
R2#
R2#copy run startup-config
R2#reload
R2>enable
R2#show version



(The security package is enabled)

Enter the following command in the CLI mode of Router0

Router>enable Router#configure terminal
Router(config)#hostname R0
R0(config)#

Defining the Hostname for all Routers and Configuring the Routers R1 and R2 for IPSec VPN tunnel

R1#configure terminal
R1(config)# access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key priya address 30.0.0.1
R1(config)#crypto ipsec transform-set R1->R2 esp-aes 256 esp-sha-hmac
R1(config)#

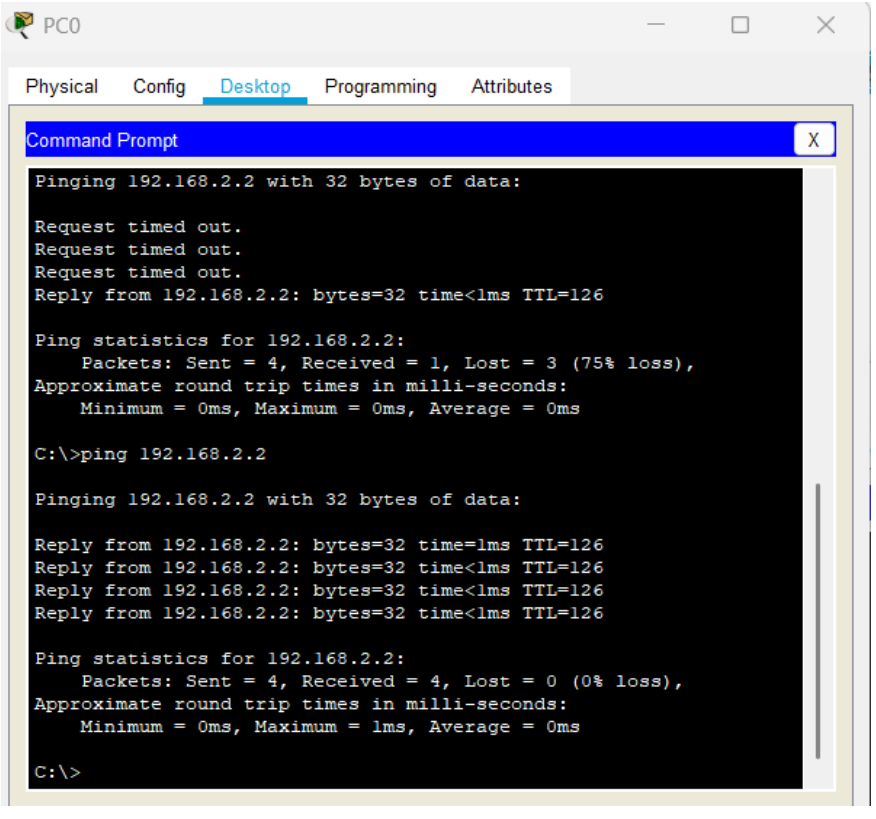
R2#
R2#configure terminal
R2(config)# access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
R2(config)#crypto isakmp policy 10
R2(config-isakmp)#encryption aes 256
R2(config-isakmp)#authentication pre-share
R2(config-isakmp)#group 5
R2(config-isakmp)#exit
R2(config)#crypto isakmp key priya address 20.0.0.1
R2(config)#crypto ipsec transform-set R2->R1 esp-aes 256 esp-sha-hmac
R2(config)#

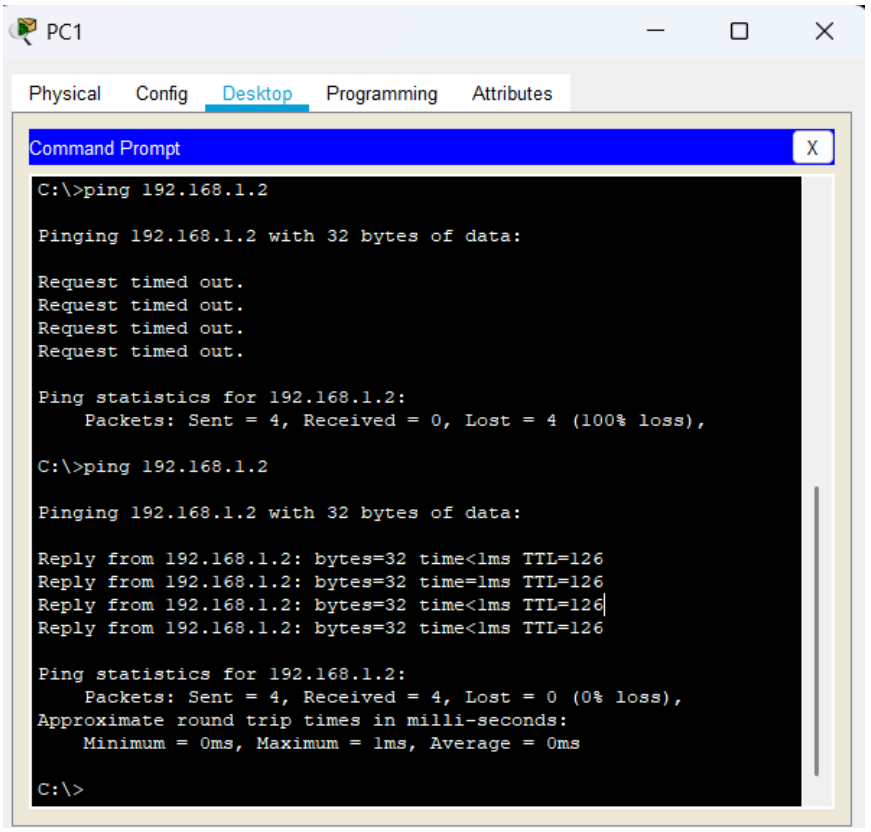
R1>enable
R1#configure terminal
R1(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
R1(config-crypto-map)#set peer 30.0.0.1
R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)#set security-association lifetime seconds 86400
R1(config-crypto-map)#set transform-set R1->R2
R1(config-crypto-map)#match address 100
R1(config-crypto-map)#exit
R1(config)#interface g0/0
R1(config-if)#crypto map IPSEC-MAP

R2>enable
R2#configure terminal
R2(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
R2(config-crypto-map)#set peer 20.0.0.1
R2(config-crypto-map)#set pfs group5
R2(config-crypto-map)#set security-association lifetime seconds 86400
R2(config-crypto-map)#set transform-set R2->R1
R2(config-crypto-map)#match address 100
R2(config-crypto-map)#exit
R2(config)#interface g0/0
R2(config-if)#crypto map IPSEC-MAP

We verify the working of the IPSec VPN tunnel using the ping command as follows:

Output : (192.168.2.2) from PC1 and then PC1(192.168.1.2) from PC2




PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>